# EDA LAB ASSESSEMENT

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1. Calculator

Code:

```
#calc
#using for loops
ch=''
while ch!=5:
    print('''
    Menu
    1.Addition
    2.Subtraction
    3.Multiplication
    4.Division
    5.Exit''')

    ch=int(input("Enter option: "))
    if ch==1:
        a=float(input("Enter first number: "))
        b=float(input("Enter second number: "))
        print(a+b)
    elif ch==2:
        a=float(input("Enter first number: "))
        b=float(input("Enter second number: "))
        print(a-b)
    elif ch==3:
        a=float(input("Enter first number: "))
        b=float(input("Enter second number: "))
        print(a*b)
    elif ch==4:
        a=float(input("Enter first number: "))
        b=float(input("Enter second number: "))
        if b==0:
            print("Cannot divide by 0")
        else:
            print(a/b)
    else:
        print("Exiting...")
```

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> & C:/Users/abhij/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/abhij/Documents/Mine/Abhijeeth's/2025 VIT college work/SEM3/EDA/EDA lab/calc.py"

    Menu
    1.Addition
    2.Subtraction
    3.Multiplication
    4.Division
    5.Exit
Enter option: 1
Enter first number: 12
Enter second number: 13
25.0

    Menu
    1.Addition
    2.Subtraction
    3.Multiplication
    4.Division
    5.Exit
Enter option: 4
Enter first number: 13
Enter second number: 0
Cannot divide by 0
```

2. largest of a set of numbers

Code:

```
#largest of a set of numbers
#using lists
l=[]
n=int(input("Enter number of terms: "))
for i in range (0,n):
    a=int(input("Enter number: "))
    l.append(a)
print("largest: ",max(l))
```

Output

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> python -u "c:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab\largest.py"
Enter number of terms: 5
Enter number: 21
Enter number: 312
Enter number: 434
Enter number: 123
Enter number: 433
largest:  434
```

3. phone book

Code:

```
#phone book
#using dictionary

dic={}
ch=1
while ch==1:
    print('''
1.Enter name and num
2.Show phone book
3.Exit''')
    ch=int(input("Enter choice: "))
    if ch==1:
        name=input("Enter name: ")
        num=int(input("Enter number: "))
        dic[name]=num
    elif ch==2:
        print(dic)
    elif ch==3:
        print("Exiting..")
```

Output

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> python -u "c:\Users\abhij\Documen
ts\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab\phonebook.py"

1.Enter name and num
2.Show phone book
3.Exit
Enter choice: 1
Enter name: John
Enter number: 9876543210

1.Enter name and num
2.Show phone book
3.Exit
Enter choice: 1
Enter name: James
Enter number: 123456789

1.Enter name and num
2.Show phone book
3.Exit
Enter choice: 2
{'John': 9876543210, 'James': 123456789}
```

4. Palindrome Check

Code:

```
text = input("Enter a word: ")
if text == text[::-1]:
    print("Palindrome")
else:
    print("Not a Palindrome")
```

Output:

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> python -u "c:\Users\abhij\Documen
ts\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab\palindrome.py"
Enter a word: radar
Palindrome
```

5. Prime number check

```
num = int(input("Enter a number: "))
if num <= 1:
    print("Not Prime")
else:
    is_prime = True

    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        print("Prime Number")
    else:
        print("Not Prime")
```

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> python -u "c:\Users\abhij\Documen
ts\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab\primenum.py"
Enter a number: 13
Prime Number
```

6. Fruit check

Code:

```
fruits = ("Apple", "Banana", "Orange", "Mango")

print("Tuple:", fruits)
print("First fruit:", fruits[0])
print("Last fruit:", fruits[-1])
print("Total fruits:", len(fruits))

search = input("Enter a fruit to search: ")

if search in fruits:
    print(search, "is present in the tuple.")
else:
    print(search, "is not present.")
```

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> python -u "c:\Users\abhij\Documen
ts\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab\fruit.py"
Tuple: ('Apple', 'Banana', 'Orange', 'Mango')
First fruit: Apple
Last fruit: Mango
Total fruits: 4
Enter a fruit to search: Banana
Banana is present in the tuple.
```

7. Set operations

Code:

```
set1 = {1, 2, 3, 4}
set2 = {3, 4, 5, 6}

print("Set 1:", set1)
print("Set 2:", set2)

print("Union:", set1 | set2)
print("Intersection:", set1 & set2)
print("Difference:", set1 - set2)
```

Output:

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> python -u "c:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab\set.py"
Set 1: {1, 2, 3, 4}
Set 2: {3, 4, 5, 6}
Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference: {1, 2}
```

8. Exception Handling with Division

Code:

```
try:
    a = int(input("Enter first number: "))
    b = int(input("Enter second number: "))
    result = a / b
    print("Result:" , result)
except ZeroDivisionError:
    print("Cannot divide by zero.")
except ValueError:
    print("Please enter valid numbers.")
```

Output:

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> python -u "c:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab\division.py"
Enter first number: 12
Enter second number: 0
Cannot divide by zero.
```

9. Factorial

Code:

```
num = int(input("Enter a number: "))
fact = 1
for i in range(1, num + 1):
    fact *= i
print("Factorial:", fact)
```

Output:

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> python -u "c:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab\factorial.py"
Enter a number: 6
Factorial: 720
```

10. Employee Salary Calculator

```python
employees = []
n = int(input("Enter number of employees: "))
for i in range(n):
    name = input("Employee Name: ")
    salary = float(input("Basic Salary: "))
    bonus = salary * 0.10
    total = salary + bonus
    employees.append({
        "Name": name,
        "Salary": salary,
        "Bonus": bonus,
        "Total": total
    })
print("\nEmployee Details")
for emp in employees:
    print("-----------------------")
    print("Name :", emp["Name"])
    print("Salary :", emp["Salary"])
    print("Bonus :", emp["Bonus"])
    print("Total :", emp["Total"])
```

```
PS C:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab> python -u "c:\Users\abhij\Documents\Mine\Abhijeeth's\2025 VIT college work\SEM3\EDA\EDA lab\tempCodeRunnerFile.py"
Enter number of employees: 3
Employee Name: James
Basic Salary: 2000
Employee Name: George
Basic Salary: 200
Employee Name: John
Basic Salary: 50000

Employee Details
-----------------------
Name : James
Salary : 2000.0
Bonus : 200.0
Total : 2200.0
-----------------------
Name : George
Salary : 200.0
Bonus : 20.0
Total : 220.0
-----------------------
Name : John
Salary : 50000.0
Bonus : 5000.0
Total : 55000.0
```